

Here's the website for the ultrafast optics group led by Oren Cohen. Pavel Sidorenko, one of his group members is the first author on the paper:

P. Sidorenko, O. Lahav, Z. Avnat, and O. Cohen, "Ptychographic reconstruction algorithm for frequency-resolved optical gating: super-resolution and supreme robustness," *Optica* 3, 1320 (2016).

<https://oren.net.technion.ac.il/ptych-frog/>

The source code is not directly linked in the paper, so you have to go to their group website to download (for free! 😊). In particular, it's apparent in their source code that the algorithm is slightly different than the step-by-step process laid out in the paper: The algorithm in the paper only updates the "object", but the correct algorithm includes another term that also updates the "probe".

Tips on usage

- You can keep the intensity autocorrelation threshold at $1/e$. And try different frequency bandwidths until the retrieval converges more consistently.

Notes to self

- Unless you blow up the frequency bandwidth to be significantly wider than the frequency marginal, even a bandwidth close to x2 the support of the frequency marginal would entail a low SNR threshold that is applied only to the outside few frequency bins. Which doesn't really do anything. So, I haven't implemented the low SNR threshold.

More notes to self

The major pitfall I fell into for all of PhD was the conversion from the fundamental to second harmonic frequency grid. I think I have it right now 🙌

The phase-matching curve given by Rick Trebino for second harmonic generation is shown below. You can see it's given as a function of the output (second-harmonic) frequency Ω , and that the conversion between the SHG and fundamental grids is $\Omega \leftrightarrow \Omega/2$. This is the correct conversion to calculate the phase-matching curve for the second harmonic pathway.

$$R(\Omega) = Q(\Omega) \frac{\Omega^3}{n_e(\Omega)} \left\{ [n_e^2(\Omega) - 1] [n_o^2(\Omega/2) - 1]^2 \right\}^2 \cdot \text{sinc}^2 \left(\frac{\Delta k(\Omega/2, \Omega/2)L}{2} \right)$$

However: The input frequency grid ($\Omega/2$) to the SHG pathway is not the same as the retrieval grid! The conversion from the spectrogram's frequency axis to the fundamental is **not** $\Omega \leftrightarrow \Omega/2$. Rather, it is simply a **translation** of the center frequency by $\Omega_0/2$:

We can derive this relation. Consider the pulse envelope:

$$A(t)e^{i\omega_0 t} + c. c.$$

It is easy to consider the case when the time delay is 0, where we have second harmonic generation:

$$(A(t)e^{i\omega_0 t} + c. c.)^2 = A(t)^2 e^{2i\omega_0 t} + 2|A(t)|^2 + A^*(t)^2 e^{-2i\omega_0 t}$$

Removing the DC offset and taking only positive frequencies, we have that the FROG signal is given by:

$$I(\omega, 0) = \mathcal{F} \{ A(t)^2 e^{2i\omega_0 t} \}(\omega)$$

Looking at this expression we can extend to other time delays:

$$I(\omega, \tau) = \mathcal{F} \{ A(t)A(t - \tau)e^{2i\omega_0 t} \}(\omega)$$

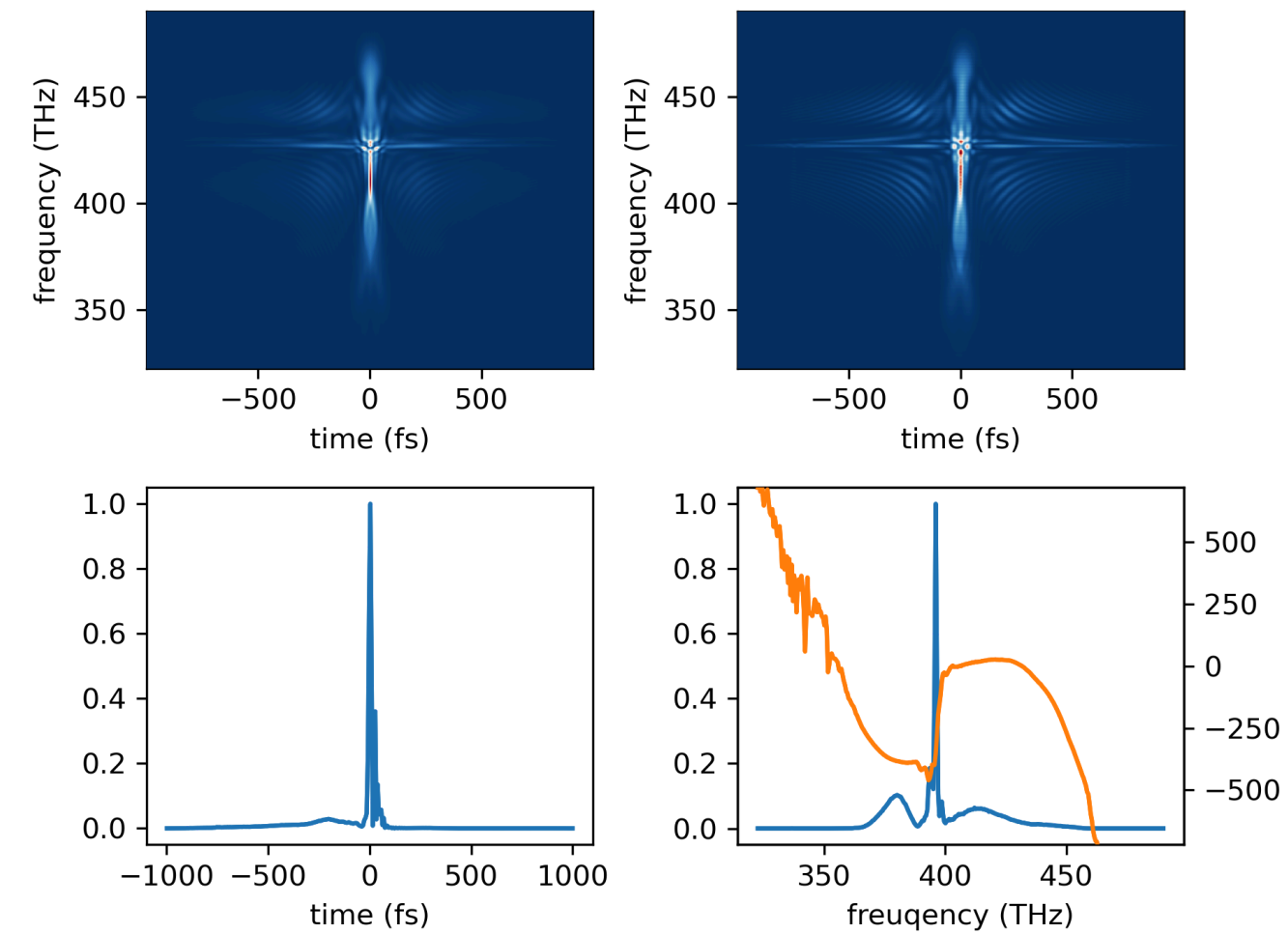
Now looking at this expression we can draw the conclusion: When setting up the retrieval for $A(t)$, if you use the frequency grid defined by your FROG spectrometer, you will retrieve the pulse on the fundamental frequency grid centered at the second harmonic $2\omega_0 t$.

So, in conclusion: the conversion is:

$$\omega_{\text{fundamental}} = \omega_{\text{FROG}} - \omega_{\text{FROG}}/2$$

Some example retrievals

Top left is re-constructed, top right is experimental



Here's the spectra compared to the experimental. They're both just normalized. The blue is the experimental.

